



Utilization of magnetic and electrostatic separation in the recycling of printed circuit boards scrap

H.M. Veit ^a  , T.R. Diehl ^a, A.P. Salami ^a, J.S. Rodrigues ^a, A.M. Bernardes ^a, J.A.S. Tenório ^b

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Abstract

The progress of the technology is directly related to the growth of production and consumption of electrical/electronics equipment, especially of personal computers. This type of equipment has a relatively short average lifetime, 2–3 years. The amount of defective or obsolete equipment has been increasing substantially; consequently its disposition and/or recycling should be studied.

In this work, printed circuit boards, which are used in personal computers, were studied in order to recover the metals in the circuit boards through mechanical processing, such as crushing, screening, as well as magnetic and electrostatic separation.

The results obtained demonstrate the feasibility of using these processes to separate metal fractions from polymers and ceramics, and that it is possible to obtain a fraction concentrated in metals containing more than 50% on average of copper, 24% of tin and 8% of lead.

Introduction

The quantity of electro/electronic equipment produced nowadays is very high and it has been constantly increasing due to the growth in the consumption of these goods by the population as well as the relatively short lifetime of this equipment.

On the basis of its high electrical conductivity, copper is the one of the basic materials of electrical engineering technology.

Bertram et al. (2002) studied the waste management system, especially the copper cycle in Europe, and they affirm that wastes from electrical and electronic equipment (WEEE) are the fastest growing waste category; this finding emphasizes the need for efficient WEEE recycling strategies.

In western Europe, 6 million ton of electric and electronic waste were generated in 1998. The amount of this waste is expected to increase by at least 3–5% per annum (Forssberg and Cui, 2003). According to Bertram et al. (2002), the generation of WEEE is around 7 kg per capita per year in Europe. In the USA, a recent study predicted that over 315 million computers would be at the end of their life by the year 2004 (Forssberg and Cui, 2003).

In Taiwan, in terms of industries producing post consumer wastes, the top three are electronics and electrical engineering industries, with 630.000 ton (43%), chemicals with 450.000 ton (31%) and metal industries with 200.000 ton (14%) (Wei, 2000).

The recycling of this scrap is still quite limited due to the heterogeneity of the materials present on the product and the complexity of the production of this equipment.

Printed circuit boards are part of this equipment and their composition is quite varied, containing polymers, ceramics and metals. The metal content is around 28% (copper: 10–20%, lead: 1–5%, nickel: 1–3%). The content of the most important materials remaining is: plastics 19%, bromine (especially as flame-retardant) 4%, glass and ceramics 49% (Ludwig et al., 2003). In addition, there are also considerable quantities of toxicologically relevant substances, such as the flame-retarding “synergist” antimony trioxide, which is suspected to be carcinogenic, as well as precious metals (in decreasing order: silver, platinum metals like palladium, gold, to a total of ≈ 0.3 –0.4%).

Aside from the special case of precious metals, copper in circuit boards is economically the most interesting metal for recovery and use as a secondary raw material because of its relatively high content compared to the corresponding content in explorable ores and its economic value. Copper recovery is also of ecological importance because of the relatively high copper content in comparison to the earth’s crust composition and the corresponding low acceptable concentration limits, concerning leachate from landfills (Ludwig et al., 2003). Nevertheless, the contents of some chemical elements are much

higher than in earth's crust, especially of some harmful chemical elements like bromine, antimony, cadmium and lead.

In addition to these inorganic elements, the following important organic compounds are found in circuit boards: isocyanates and phosgene from polyurethanes, acrylic and phenolic resins, epoxides and phenols, such as chip glues (Ludwig et al., 2003). Therefore, the circuit boards cannot be considered inert materials and are not suited for direct disposal in landfills.

The existing processes of recycling printed circuit boards use pyrometallurgical (Felix and Riet, 1994, Schichang et al., 1994, Bernardes et al., 1997, Szczygiel et al., 1998) or hydrometallurgical methods (Gloe et al., 1990, Pozzo et al., 1991, Hoffmann, 1992, Shibata and Matsumoto, 1999, Marca et al., 2002), which generate atmospheric pollution through the release of dioxins and furans (Menad et al., 1998) or high volumes of effluents.

Some authors are using mechanical processing (Zhang and Forssberg, 1997, Tenório et al., 1997, Noakes, 1999, Veit et al., 2002a) as an alternative to concentrate the metals in a fraction and the polymers and ceramics in another. The metal concentrate fraction can be sent to electrochemistry processes (Brown, 1992, Scott et al., 1997, Fornari and Abbruzzese, 1999, Kékesi et al., 2000, Brandon et al., 2002, Ubaldini et al., 2003) in order to separate the metals among themselves.

This work intends to study magnetic and electrostatic separation processes in order to obtain fractions with higher metals concentration, especially copper, from printed circuit boards scrap, previously comminuted and separated by size.

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Experimental

In this work, mechanical processing was used as an alternative to processes that are usually used (thermal and hydrometallurgical processes) to the recycling of copper. This was done in order to concentrate metals in a fraction and polymeric and ceramic

materials in another, without great amount of energy consumption and atmospheric pollution generation, case of thermal processes, or great amount of effluents generation, case of hydrometallurgical processes.

The steps of mechanical processing ...

Magnetic separation

The results are presented in Fig. 2, which shows quantities (mass) of magnetic material and non-magnetic material in each fraction after magnetic separation for whole printed circuit board and electronic components. Fig. 3 presents the contents of Fe, Ni, Cu and Pb after chemical analysis. The metals that should be analyzed were chosen as follows: in Fe and Ni cases, because they are ferromagnetic, they should be present after a magnetic separation; in the case of Cu, since this is the most ...

Discussion

It was expected that the fractions that were magnetically separated should have high concentrations of Fe, proving the efficiency of this type of separation.

Although the amount of magnetic material present in printed circuit boards is small, it is interesting to separate it previously, in order to obtain conductive fractions with higher copper content. The largest amount of magnetic material in whole printed circuit boards was obtained for the fraction F3: 54 g in 2 kg of samples. In electronic ...

Conclusions

The use of magnetic separation and electrostatic separation was efficient to obtain fractions with relatively high concentrations of metals starting from scrap of printed circuit boards. The fractions magnetically separated reached 43% of iron on average in whole printed circuit boards scrap (PCB) and 46% of iron in electronic components scrap (EC). The fractions electrostatically separated presented, for PCB, medium concentrations of 50% copper, 25% tin and 7% lead. For EC the concentrations ...

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